

A. Innovation Summary

Project Title: Smart Beneficiary Mapping System (SBMS)

Problem Statement: Millions of eligible Indian citizens remain excluded from 4,700+ Central and State welfare schemes due to severe information asymmetry, fragmented departmental websites, complex eligibility rules, and predatory middlemen.

Proposed Solution: SBMS is a privacy-first Digital Public Infrastructure that autonomously maps citizens to welfare entitlements with zero-click proactive matching and direct official portal redirection.

Use of Artificial Intelligence: SBMS integrates 100% on-premise local AI—Llama 3 for multilingual conversational assistance, Qwen 2.5-Coder for structured rule evaluation, Qwen 2.5-VL for multimodal document OCR, and pgvector HNSW dense embeddings (all-MiniLM-L6-v2) for sub-second semantic retrieval.

Innovation / Novelty: Replaces tedious 30-step questionnaires with real-time vector matching, provides an AI Document Vault with automated PII scrubbing (masking Aadhaar/PAN), verifies certificates before application (classifying schemes as "Eligible & Verified" vs "Documents Pending"), and deep-links directly to authentic Ministry portals.

Target Users / Beneficiaries: Smallholder farmers, low-income families (BPL/EWS), female entrepreneurs, students, senior citizens, and persons with disabilities across all 36 States and Union Territories, alongside government welfare officers.

Expected Impact: Eliminates middleman leakage to ensure 100% Direct Benefit Transfer (DBT) delivery, enables paperless digital governance, and accelerates socio-economic inclusion.

Current Development Stage: Working Prototype with 4,725 live indexed government schemes.